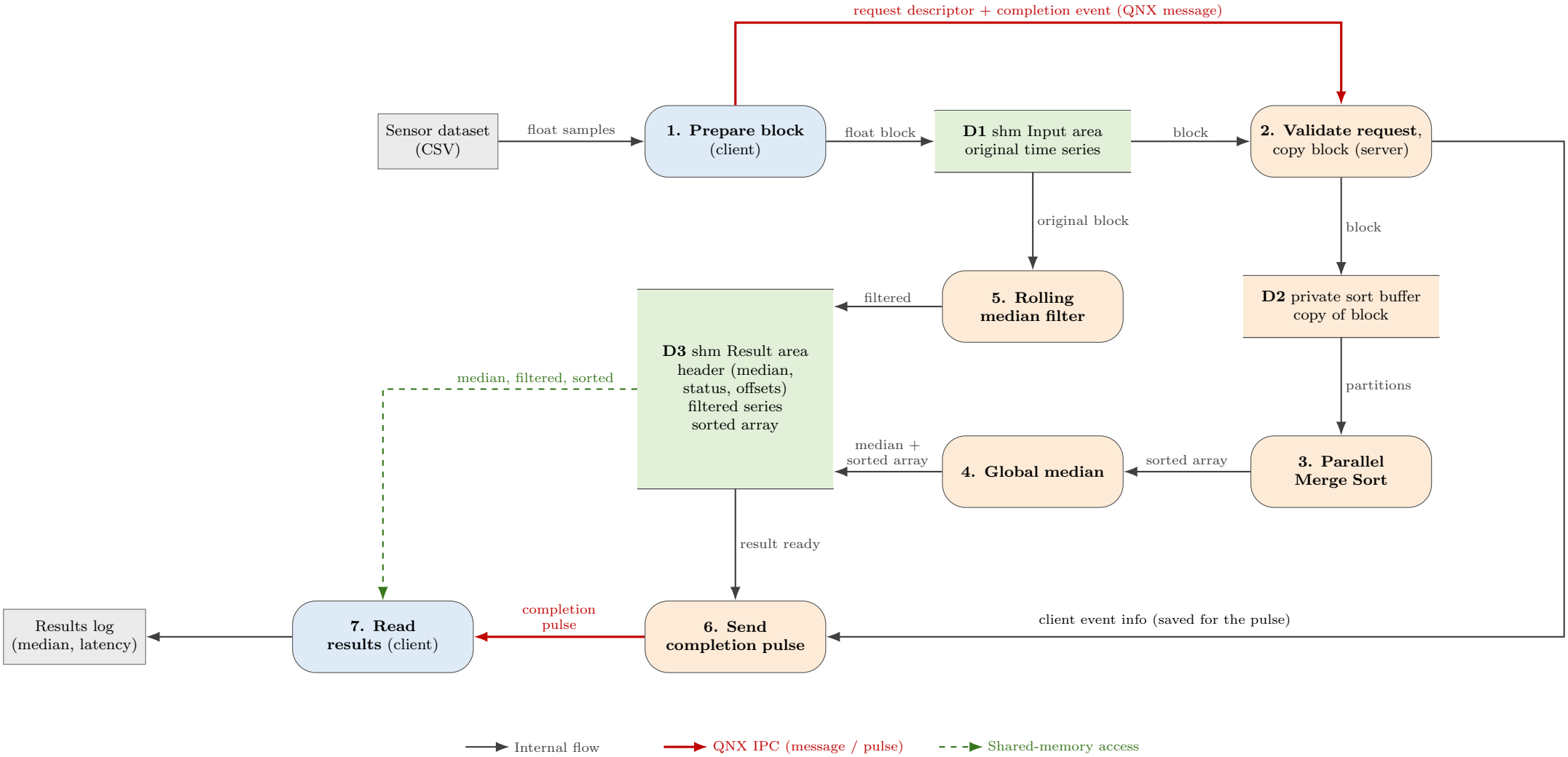
request descriptor + completion event (QNX message)
Sensor dataset (CSV)
float samples
1. Prepare block (client)
float block
D1 shm Input area original time series
block
2. Validate request, copy block (server)
original block
block
5. Rolling median filter
filtered
D2 private sort buffer copy of block
D3 shm Result area header (median, status, offsets) filtered series sorted array
median, filtered, sorted
partitions
median + sorted array
4. Global median
sorted array
3. Parallel Merge Sort
result ready
Results log (median, latency)
7. Read results (client)
completion pulse
6. Send completion pulse
client event info (saved for the pulse)
Internal flow
QNX IPC (message / pulse)
Shared-memory access